

Adding and Subtracting Fractions with Like Denominators

Grades 4–5

Adding, subtracting, and finding a missing numerator — all with like denominators

The big idea. When two fractions have the **same denominator**, the pieces are the *same size*, so you can just count them.

- Add or subtract the **numerators** (the counts of pieces).
- Keep the **denominator** (the size of one piece) exactly as it is.
- Shade or cross out on the bar model first, then write the number sentence.
- Write your final answer in simplest form when it can be simplified.

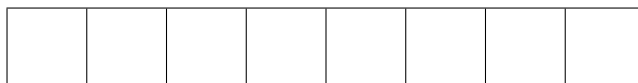
1. Each bar below is cut into **5** equal parts. Shade $\frac{1}{5}$ on the first bar and $\frac{3}{5}$ more on the same bar.



How much of the bar is shaded in all? $\frac{1}{5} + \frac{3}{5} =$

Answer: _____

2. This bar is cut into **8** equal parts. Shade $\frac{2}{8}$, then shade $\frac{5}{8}$ more.



Write the sum. $\frac{2}{8} + \frac{5}{8} =$

Answer: _____

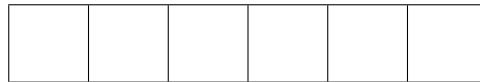
3. Start by shading $\frac{7}{9}$ of this bar, then cross out $\frac{4}{9}$ of it.



How much is still shaded? Write the difference, then simplify it. $\frac{7}{9} - \frac{4}{9} =$

Answer: _____

4. A bar is cut into 6 equal parts and $\frac{5}{6}$ of it is shaded. Cross out $\frac{1}{6}$.



Write the difference in simplest form. $\frac{5}{6} - \frac{1}{6} =$

Answer: _____

5. Two bars, each cut into 8 equal parts, are shown side by side.



Shade $\frac{5}{8}$ on the first bar and $\frac{4}{8}$ on the second. Add them. The sum is bigger than one whole bar — write it as a fraction, and also say how many whole bars and how many eighths that is.

Answer: _____

6. **Missing numerator.** Fill in the blank so the sentence is true. Use the ninths bar to help you count up from $\frac{2}{9}$ to $\frac{7}{9}$.



$$\frac{2}{9} + \underline{\hspace{2cm}} = \frac{7}{9}$$

Answer: _____

7. Story problem. A chocolate bar is cut into **12** equal pieces. Lena has $\frac{11}{12}$ of the bar left over from yesterday. Today she eats $\frac{5}{12}$ of the bar. What fraction of the bar is left now? Give the answer in simplest form.

Answer: _____

8. Three parts of one whole. Three friends shade parts of the same sixths bar: Ana shades $\frac{1}{6}$, Ben shades $\frac{2}{6}$, and Cruz shades $\frac{2}{6}$.



What fraction of the bar is shaded altogether? Is the bar completely full? Explain using the bar.

Answer: _____

9. Work backwards. A jug is marked in eighths. Someone poured out $\frac{3}{8}$ of the jug, and $\frac{4}{8}$ of the jug is still there.

What fraction of the jug was there before the pouring? Write the equation you used:

$$\underline{\hspace{2cm}} - \frac{3}{8} = \frac{4}{8}$$

Answer: _____

10. Find and fix the mistake. Jae wrote this on a test:

$$\frac{3}{7} + \frac{2}{7} = \frac{5}{14}$$

Use a sevenths bar to show why Jae's answer cannot be right, explain in one sentence what Jae did wrong, and write the correct sum.

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Answer: _____